

☒ Linux Software Firewall, Reverse Proxy, UTM & Load Balancing

☒ 1. Linux Software Firewalls (ClearOS & pfSense)

Linux software firewalls offer packet filtering, NAT, and routing capabilities via tools or distributions like:

☒ ClearOS

- A CentOS-based Linux distribution
- Designed as a UTM (Unified Threat Management) solution
- Web-based GUI
- Modules for:

- Firewall & Intrusion Detection
- VPN (IPSec/OpenVPN)
- Web proxy
- Email filtering
- Bandwidth management

☒ pfSense

- Based on **FreeBSD**, not Linux, but widely used as a **software firewall/router**

- GUI-driven, enterprise-grade features:
 - Stateful packet filtering
 - NAT & Port forwarding
 - VPN: OpenVPN, IPSec
 - Traffic shaping
 - IDS/IPS (Snort, Suricata)
 - Package manager (Squid, pfBlockerNG, etc.)
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☒ Common Features in Both:

Feature	Description
Firewall Rules	Define what traffic is allowed/blocked
NAT	Translate internal IPs for internet access
IDS/IPS	Detect or block suspicious activity
VPN	Secure remote access
Web Proxy	Cache and control web traffic
Logging/Monitoring	See traffic trends and logs

☒ 2. Nginx & Squid Reverse Proxy

☒ What is a Reverse Proxy?

A **reverse proxy** forwards **client requests to backend servers** and returns the response as if it came from the proxy.

📦 Nginx

- High-performance reverse proxy and web server
- Load balancing + caching + SSL termination

📦 Example Nginx reverse proxy block:

nginx

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```
server { listen 80; server_name app.example.com; location / { proxy_pass http://127.0.0.1:5000; } }
```

❏ Squid

- Traditionally a **forward proxy**, but supports **reverse proxy mode**
- Mainly for **web acceleration, caching, and content filtering**

❏ Reverse proxy example:

bash

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```
http_port 80 accel defaultsite=www.backend.com cache_peer 192.168.1.100 parent 80 0 no-query originserver name=websrv
```

⌘ Comparison: Nginx vs Squid

Feature	Nginx	Squid
Focus	Web + App Proxy	Web Caching/Filtering
Reverse Proxy	Native support	Configured with accel
SSL Termination	Supported	Limited
Performance	Very high	Good for caching

⌘ 3. Unified Threat Management (UTM)

UTM is an all-in-one security solution that combines:

- Firewall

- IDS/IPS
- VPN Gateway
- Web Content Filtering
- Antivirus Gateway
- Email security (spam/phishing)

☒ Examples:

- ClearOS
- pfSense with plugins (Snort, ClamAV)

- Sophos UTM
- FortiGate

☒ Benefits:

- Centralized security management
- Lower cost for SMEs
- Easier maintenance

☒ 4. Server Load Balancing

⌘ Definition:

Distributes incoming client requests across multiple servers to optimize:

- Speed
 - Redundancy
 - High availability
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⌘ Common Load Balancing Techniques:

Method	Description
Round Robin	Rotates requests among servers
Least Connections	Chooses server with fewest active connections

IP Hash	Directs a client IP to the same server
Health Checks	Only forwards to healthy servers

☒ Tools:

- Nginx: Load balancer + reverse proxy
- HAProxy: Advanced TCP/HTTP LB
- pfSense: Built-in load balancer
- AWS ELB, Kubernetes Services

📄 Nginx Load Balancing Example:

nginx

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```
upstream backend { server app1.example.com; server app2.example.com; } server { location / { proxy_pass http://backend; } }
```